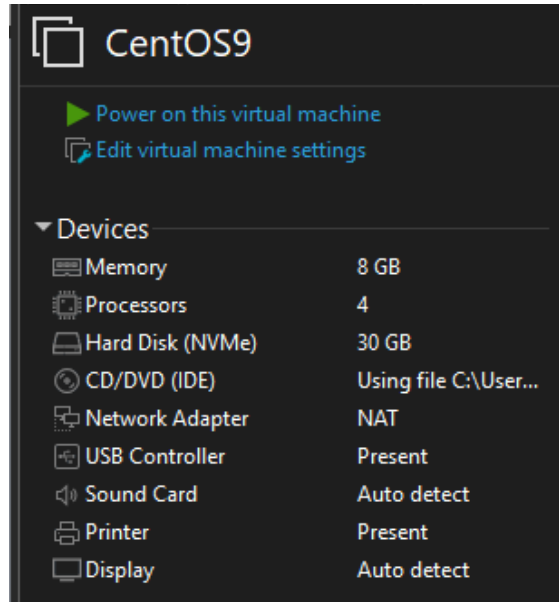


Linux Lab 4

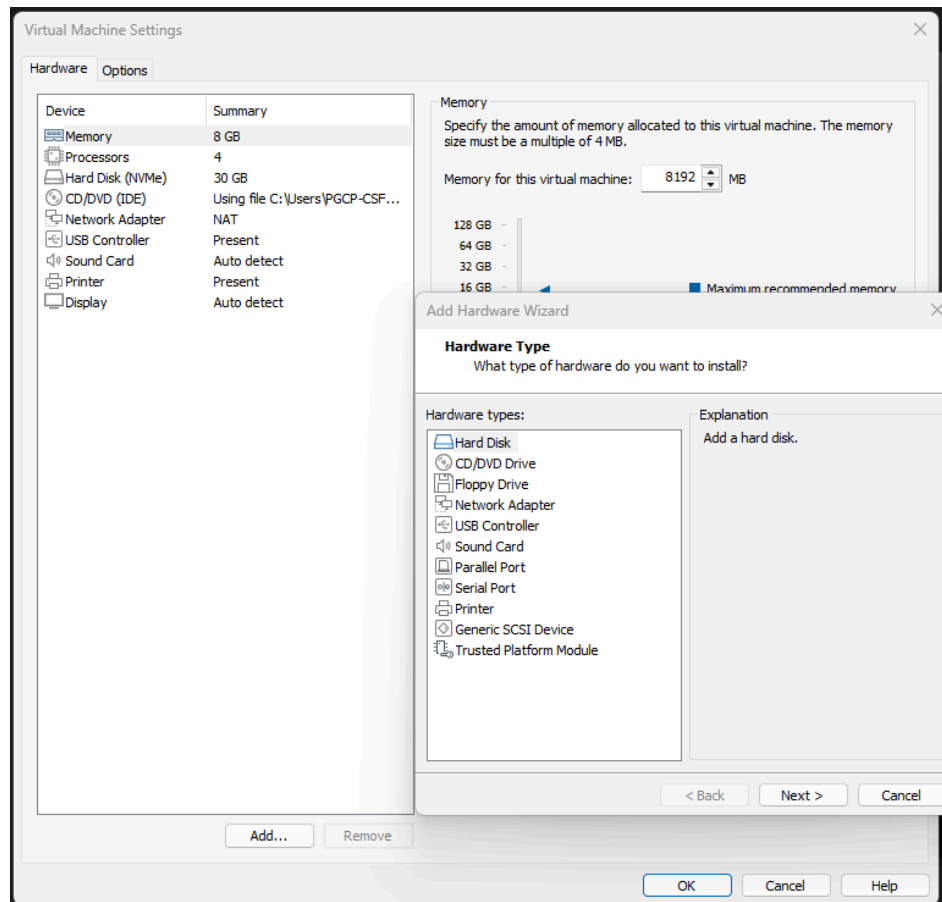
By Jai Pandey (260840140011)

Question 1

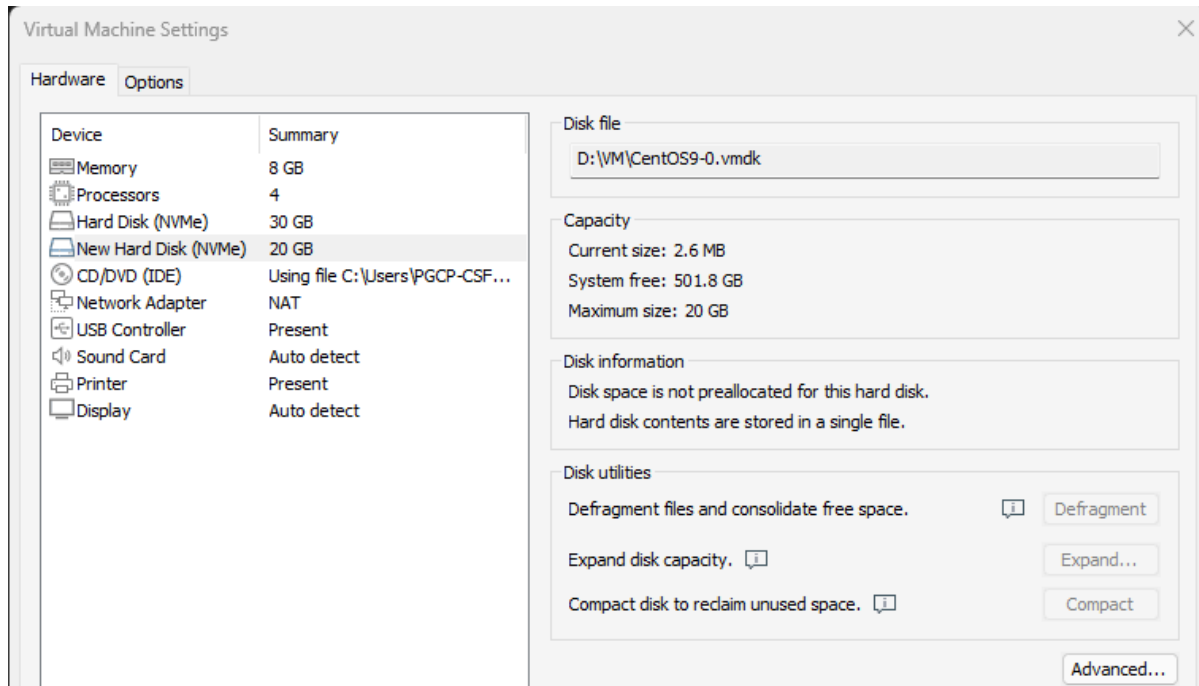
1. Editing the virtual machine settings



2. Adding a new Virtual Hard Disk



3. Hard Disk > NVMe > Create New > Store virtual disk as single file> Save Virtual Hard disk



4. Seeing storage info using fdisk

```
[root@localhost csf]# fdisk -l
Disk /dev/nvme0n1: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0d36fe25

Device            Boot    Start        End    Sectors    Size Id Type
/dev/nvme0n1p1    *                2048    4196351    4194304     2G 83 Linux
/dev/nvme0n1p2          4196352    46139391    41943040    20G 83 Linux
/dev/nvme0n1p3          46139392    62914559    16775168     8G 82 Linux swap / Solaris

Disk /dev/nvme0n2: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
[root@localhost csf]#
```

5. Creating Partition 1 (/mnt/data)

```
Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (1-4, default 1): 1
First sector (2048-41943039, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-41943039, default 41943039): +15G

Created a new partition 1 of type 'Linux' and of size 15 GiB.
```

6. Creating Partition 2 (/database)

```
Command (m for help): n
Partition type
  p   primary (1 primary, 0 extended, 3 free)
  e   extended (container for logical partitions)
Select (default p): p
Partition number (2-4, default 2): 2
First sector (31459328-41943039, default 31459328):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-41943039, default 41943039): +5G
Value out of range.
Last sector, +/-sectors or +/-size{K,M,G,T,P} (31459328-41943039, default 41943039):

Created a new partition 2 of type 'Linux' and of size 5 GiB.

Command (m for help): p
Disk /dev/nvme0n2: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x31b9c1e5

Device          Boot      Start          End      Sectors  Size Id Type
/dev/nvme0n2p1                2048 31459327 31457280    15G 83 Linux
/dev/nvme0n2p2   31459328 41943039 10483712     5G 83 Linux
```

7. Saving changes

```
Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@localhost csf]#
```

8. Adding filesystem for partition 1

```
[root@localhost csf]# mkfs.ext4 /dev/nvme0n2p1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: 206e28d3-a8b9-4cbd-9db3-0a0759be54fc
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost csf]#
```

9. Adding filesystem for partition 2

```
[root@localhost csf]# mkfs.ext4 /dev/nvme0n2p1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: 206e28d3-a8b9-4cbd-9db3-0a0759be54fc
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done
```

10. Mounting both partitions

```
[root@localhost /]# mount /dev/nvme0n2p1
[root@localhost /]# mount /dev/nvme0n2p2
[root@localhost /]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	3.8G	0	3.8G	0%	/dev
tmpfs	3.8G	0	3.8G	0%	/dev/shm
tmpfs	1.5G	9.8M	1.5G	1%	/run
/dev/nvme0n1p2	20G	6.6G	12G	36%	/
tmpfs	765M	96K	765M	1%	/run/user/1000
/dev/nvme0n1p1	2.0G	449M	1.5G	23%	/boot
/dev/nvme0n2p1	15G	24K	14G	1%	/mnt/data
/dev/nvme0n2p2	4.9G	92K	4.6G	1%	/database

```
[root@localhost /]#
```

Removing the partitions

1. Unmounting both the partitions

```
[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        3.8G     0  3.8G   0% /dev
tmpfs           3.8G     0  3.8G   0% /dev/shm
tmpfs           1.5G   9.8M  1.5G   1% /run
/dev/nvme0n1p2  20G   6.6G  12G   36% /
tmpfs           765M   96K  765M   1% /run/user/1000
/dev/nvme0n1p1  2.0G  449M  1.5G  23% /boot
/dev/nvme0n2p1   15G   24K   14G   1% /mnt/data
/dev/nvme0n2p2  4.9G   92K   4.6G   1% /database
[root@localhost ~]# umount /dev/nvme0n2p1
[root@localhost ~]# umount /dev/nvme0n2p2
[root@localhost ~]# df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        3.8G     0  3.8G   0% /dev
tmpfs           3.8G     0  3.8G   0% /dev/shm
tmpfs           1.5G   9.8M  1.5G   1% /run
/dev/nvme0n1p2  20G   6.6G  12G   36% /
tmpfs           765M   96K  765M   1% /run/user/1000
/dev/nvme0n1p1  2.0G  449M  1.5G  23% /boot
[root@localhost ~]#
```

2. Removing entry from /etc/fstab using VI

```
csf@localhost:~ — vim /etc/fstab

#
# /etc/fstab
# Created by anaconda on Thu Aug 27 12:30:19 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=0c574d4b-0a49-4ebb-a30a-853d6c1c18f8 / ext4 defaults 1 1
UUID=ddf8a755-c1b0-4359-a63a-5f662e908a69 /boot xfs defaults 0 0
UUID=08c39f26-ce60-4af0-887c-76e84e99fd6a none swap defaults 0 0
~
~
~
~
~
~
~
```

3. Deleting partition entry from partition table using fdisk

```
[root@localhost /]# fdisk /dev/nvme0n2

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): d
Partition number (1,2, default 2):
Command (m for help): d
Partition number (1,2, default 2): 1

Partition 1 has been deleted.


Command (m for help): d
Selected partition 2
Partition 2 has been deleted.


Command (m for help): 2
2: unknown command


Command (m for help): w

The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.

[root@localhost /]#
```

4. Verify removed partitions

```
[root@localhost /]# fdisk -l
Disk /dev/nvme0n1: 30 GiB, 32212254720 bytes, 62914560 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x0d36fe25

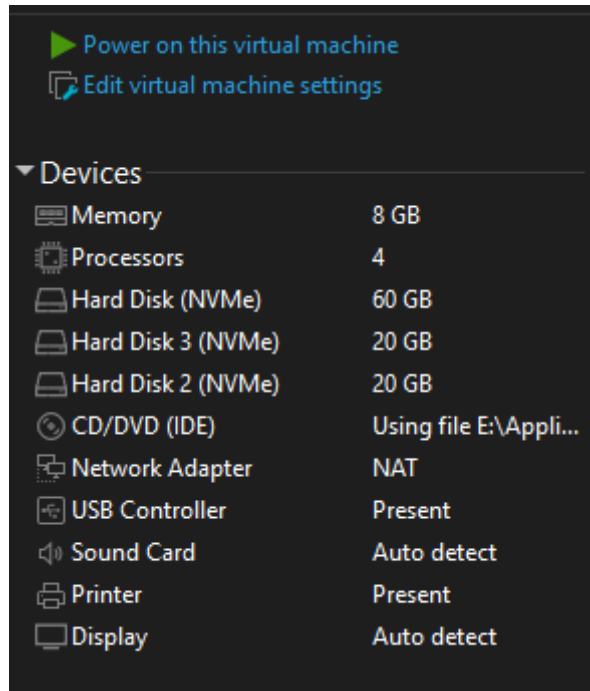
Device            Boot    Start      End  Sectors  Size Id Type
/dev/nvme0n1p1    *           2048  4196351  4194304    2G 83 Linux
/dev/nvme0n1p2             4196352  46139391  41943040   20G 83 Linux
/dev/nvme0n1p3             46139392  62914559  16775168    8G 82 Linux swap / Solaris

Disk /dev/nvme0n2: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x31b9c1e5
[root@localhost /]#
```

5. Restated the VM.

Question 2

1. Adding 2, 20GB external HDD



2. Viewing all attached disks using fdisk

```
[root@localhost csf]# fdisk -l
Disk /dev/nvme0n1: 60 GiB, 64424509440 bytes, 125829120 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x192dea6f

Device            Boot    Start        End    Sectors    Size Id Type
/dev/nvme0n1p1    *           2048    4196351    4194304     2G 83 Linux
/dev/nvme0n1p2              4196352    46139391    41943040    20G 83 Linux
/dev/nvme0n1p3              46139392    54527999     8388608     4G 82 Linux swap / Solaris

Disk /dev/nvme0n2: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/nvme0n3: 20 GiB, 21474836480 bytes, 41943040 sectors
Disk model: VMware Virtual NVMe Disk
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
[root@localhost csf]#
```

3. Creating a LINUX LVM partition on HDD - 1

```
[root@localhost csf]# fdisk /dev/nvme0n2
```

```
Welcome to fdisk (util-linux 2.37.4).
```

```
Changes will remain in memory only, until you decide to write them.  
Be careful before using the write command.
```

```
Command (m for help): p
```

```
Disk /dev/nvme0n2: 20 GiB, 21474836480 bytes, 41943040 sectors
```

```
Disk model: VMware Virtual NVMe Disk
```

```
Units: sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
```

```
Disk identifier: 0x4ad23a78
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/nvme0n2p1		2048	41943039	41940992	20G	83	Linux

```
Command (m for help): t
```

```
Selected partition 1
```

```
Hex code or alias (type L to list all): 8E
```

```
Changed type of partition 'Linux' to 'Linux LVM'.
```

```
Command (m for help): w
```

```
The partition table has been altered.
```

```
Calling ioctl() to re-read partition table.
```

```
Syncing disks.
```

```
[root@localhost csf]# partprobe
```

```
Warning: Unable to open /dev/sr0 read-write (Read-only file system).
```

```
[root@localhost csf]# pvcreate /dev/nvme0n2p1
```

```
Physical volume "/dev/nvme0n2p1" successfully created.
```

```
Creating devices file /etc/lvm/devices/system.devices
```

```
[root@localhost csf]# pvdisplay
```

```
"/dev/nvme0n2p1" is a new physical volume of "<20.00 GiB"
```

```
--- NEW Physical volume ---
```

```
PV Name                /dev/nvme0n2p1
```

```
VG Name
```

```
PV Size                <20.00 GiB
```

```
Allocatable            NO
```

```
PE Size                0
```

```
Total PE              0
```

```
Free PE                0
```

```
Allocated PE          0
```

```
PV UUID                R01y9h-Hb3c-79r8-6Ct7-efpB-hQ4N-RbenpK
```

```
[root@localhost csf]#
```

4. Creating a LINUX LVM partition on HDD - 2

```
[root@localhost csf]# fdisk /dev/nvme0n3
```

```
Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.
```

```
Command (m for help): p
```

```
Disk /dev/nvme0n3: 20 GiB, 21474836480 bytes, 41943040 sectors
```

```
Disk model: VMware Virtual NVMe Disk
```

```
Units: sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
```

```
Disk identifier: 0xb0fe9318
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/nvme0n3p1		2048	41943039	41940992	20G	83	Linux

```
Command (m for help): t
```

```
Selected partition 1
```

```
Hex code or alias (type L to list all): 8E
```

```
Changed type of partition 'Linux' to 'Linux LVM'.
```

```
Command (m for help): w
```

```
The partition table has been altered.
```

```
Calling ioctl() to re-read partition table.
```

```
Syncing disks.
```

5. Verifying changes in fdisk

```
[root@localhost csf]# fdisk -l
```

```
Disk /dev/nvme0n1: 60 GiB, 64424509440 bytes, 125829120 sectors
```

```
Disk model: VMware Virtual NVMe Disk
```

```
Units: sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
```

```
Disk identifier: 0x192dea6f
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/nvme0n1p1	*	2048	4196351	4194304	2G	83	Linux
/dev/nvme0n1p2		4196352	46139391	41943040	20G	83	Linux
/dev/nvme0n1p3		46139392	54527999	8388608	4G	82	Linux swap / Solaris

```
Disk /dev/nvme0n2: 20 GiB, 21474836480 bytes, 41943040 sectors
```

```
Disk model: VMware Virtual NVMe Disk
```

```
Units: sectors of 1 * 512 = 512 bytes
```

```
Sector size (logical/physical): 512 bytes / 512 bytes
```

```
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
```

```
Disk identifier: 0x4ad23a78
```

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/nvme0n2p1		2048	41943039	41940992	20G	8e	Linux LVM

```
Disk /dev/nvme0n3: 20 GiB, 21474836480 bytes, 41943040 sectors
```

Disk /dev/nvme0n3: 20 GiB, 21474836480 bytes, 41943040 sectors

Disk model: VMware Virtual NVMe Disk

Units: sectors of 1 * 512 = 512 bytes

Sector size (logical/physical): 512 bytes / 512 bytes

I/O size (minimum/optimal): 512 bytes / 512 bytes

Disklabel type: dos

Disk identifier: 0xb0fe9318

Device	Boot	Start	End	Sectors	Size	Id	Type
/dev/nvme0n3p1		2048	41943039	41940992	20G	8e	Linux LVM

[root@localhost csf]#

6. Created and verified 2 physical volumes for both HDD

```
[root@localhost csf]# pvs
  PV          VG Fmt  Attr PSize   PFree
  /dev/nvme0n2p1    lvm2 ---  <20.00g <20.00g
  /dev/nvme0n3p1    lvm2 ---  <20.00g <20.00g
[root@localhost csf]#
```

7. Created and verified one volume group with both physical volumes.

```
[root@localhost csf]# vgcreate vg00 /dev/nvme0n2p1 /dev/nvme0n3p1
Volume group "vg00" successfully created
[root@localhost csf]# vgs
  VG   #PV #LV #SN Attr   VSize   VFree
  vg00   2   0   0 wz--n- 39.99g 39.99g
[root@localhost csf]#
```

8. Creating 2 LVMs

```
[root@localhost csf]# lvcreate -L +10G -n lv00 vg00
Logical volume "lv00" created.
[root@localhost csf]# lvcreate -L +15G -n lv01 vg00
Logical volume "lv01" created.
[root@localhost csf]# lvs
  LV   VG   Attr      LSize   Pool Origin Data%  Meta%   Move Log Cpy%Sync Convert
  lv00 vg00 -wi-a----- 10.00g
  lv01 vg00 -wi-a----- 15.00g
[root@localhost csf]#
```

9. Formatting both LVMs as EXT4 Files system

```
[root@localhost csf]# mkfs.ext4 /dev/vg00/lv00
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: 103cb54f-0db1-4183-a40c-4cafa0a527fb
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost csf]# mkfs.ext4 /dev/vg00/lv01
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 3932160 4k blocks and 983040 inodes
Filesystem UUID: d226fdc2-e819-4029-9123-c0f92e135c13
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost csf]#
```

10. Created 2 directories to mount LVMs

11. Edited /etc/fstab to include both LVMs & reloading the daemon

```
#
# /etc/fstab
# Created by anaconda on Mon Aug 31 14:29:33 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=efaed181-59c3-4305-bf35-ba87f338b898 / xfs defaults 0 0
UUID=f849f8e1-5e4a-4a68-8ff6-5f33e7329192 /boot xfs defaults 0 0
UUID=42180408-0fd8-4405-8a89-674c3161d9d9 none swap defaults 0 0
/dev/vg00/lv00 /mnt/linux ext4 defaults 0 0
/dev/vg00/lv01 /mnt/win ext4 defaults 0 0
```

12. Verifying both LVMs are mounted

```
[root@localhost /]# df -h
Filesystem                Size      Used Avail Use% Mounted on
devtmpfs                  3.8G         0  3.8G   0% /dev
tmpfs                     3.8G         0  3.8G   0% /dev/shm
tmpfs                     1.5G    9.8M    1.5G   1% /run
/dev/nvme0n1p2             20G    4.9G    16G  25% /
/dev/nvme0n1p1             2.0G   448M    1.6G  23% /boot
tmpfs                     765M    100K    765M   1% /run/user/1000
/dev/sr0                   15G     15G         0 100% /run/media/csf/CentOS-Stream-9-BaseOS-x86_64
/dev/mapper/vg00-lv00      9.8G     24K    9.3G   1% /mnt/linux
/dev/mapper/vg00-lv01      15G     24K    14G   1% /mnt/win
[root@localhost /]#
```

13. Renaming the incorrectly named Volume Group

```
[root@localhost /]# vgrename vg00 csf
Volume group "vg00" successfully renamed to "csf"
[root@localhost /]# vgs
VG   #PV #LV #SN Attr   VSize  VFree
csf   2   2   0 wz--n- 39.99g 14.99g
```

14. Extended the Volume Group CSF by adding the newly created Physical Volume (20GB)

```
[root@localhost /]# vgextend csf /dev/nvme0n4p1
Volume group "csf" successfully extended
[root@localhost /]#
```

15. Extended the first LVM from 10GB to 25GB by adding 15 GB

```
[root@localhost /]# lvextend /dev/csf/lv00 -L +15G
Size of logical volume csf/lv00 changed from 10.00 GiB (2560 extents)
to 25.00 GiB (6400 extents).
Logical volume csf/lv00 successfully resized.
[root@localhost /]#
```